MAR-ACC-45: Arista Access Node Optical Transport Failure SOP

Node: MAR-ACC-45 | **Vendor:** Arista | **Severity:** Minor | **Alarm Code:** ALARM_6254

Description

The MAR-ACC-45 node, situated at the Marathahalli Core site and utilizing Arista's Access role, experiences Optical Transport Failure (SFP/XFP degradation, fiber cuts, Rx/Tx power loss) as a result of the complex interplay between the underlying optical transmission protocols and the hardware infrastructure. The issue arises due to the degradation of the SFP/XFP modules, which are responsible for transmitting and receiving optical signals over the fiber optic cables. When the SFP/XFP modules' receive or transmit power levels drop below a certain threshold, the node detects this anomaly and triggers an alarm (ALARM_6254) of severity Minor. This failure domain is further exacerbated by the vendor-specific CLI dialect, which necessitates adherence to specific diagnostic and resolution commands to ensure accurate troubleshooting and resolution.

Pre-requisite Checks

- Verify the site location at Marathahalli Core.
- Ensure all remote hands are dispatched to the affected site.
- Check the status of redundant links to ensure no additional failures are occurring.
- Verify power and environmental conditions at the site.

Potential Root Causes

Cause	Probability
SFP/XFP Module Degradation	High
Fiber Cut or Damage	Medium
Rx/Tx Power Level Issues	Low

Diagnostic Commands & Expected Outputs

```
Command: show interfaces e1/1/1 detail, Expected Output: Interface: e1/1/1, Status: UP, SFP/XFP Details: Optical Signal Strength: -12 dBm, Description: This command isolates the optical signal strength on the E1/1/1 interface, which will help identify if the issue is related to SFP/XFP degradation or fiber cuts. The expected output indicates a strong optical signal, but any deviation from this may indicate a problem.

Command: show environment temperature detail, Expected Output: Temperature: 25.0°C ± 5.0°C, Humidity: 60% ± 10%, Description: This command verifies the environmental conditions at the site, which may be contributing to the Optical Transport Failure. Any extreme
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temperatures or humidity levels may be causing the SFP/XFP modules to
malfunction.
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Resolution Steps

Step 1: Step: Replace SFP/XFP Module, Command: install switch e1/1/1 replace-sw-module sfp-1g-sfp-sx, Description: This command replaces the existing SFP/XFP module with a new one. Ensure to verify the new module's details and status after the replacement.

Step 2: Step: Verify Power Levels, Command: show interfaces e1/1/1 detail, Description: After replacing the SFP/XFP module, run this command to verify that the optical signal strength has returned to normal levels.

Post-Check Verification

```
Command: show interfaces e1/1/1 detail, Expected Output: Interface:
e1/1/1, Status: UP, SFP/XFP Details: Optical Signal Strength: -12 dBm,
Description: Verify that the optical signal strength is within normal
limits and the interface status is UP.
```

Roll-back Procedures

- Step: Restore Original SFP/XFP Module, Command: install switch e1/1/1 restore-sw-module sfp-1g-sfp-sx, Description: If the replacement of the SFP/XFP module fails, restore the original module to revert the changes and maintain node integrity.

- Step: Verify Original State, Command: show interfaces e1/1/1 detail, Description: Verify that the node has returned to its original state, with the original SFP/XFP module in place and the interface status UP.